

A New Strategy of Cost-Free Learning in the Class Imbalance Problem

Xiaowan Zhang and Bao-Gang Hu, *Senior Member, IEEE*

Abstract—In this work, we define cost-free learning (CFL) formally in comparison with cost-sensitive learning (CSL). The main difference between them is that a CFL approach seeks optimal classification results without requiring any cost information, even in the class imbalance problem. In fact, several CFL approaches exist in the related studies, such as sampling and some criteria-based approaches. However, to our best knowledge, none of the existing CFL and CSL approaches are able to process the abstaining classifications properly when no information is given about errors and rejects. Based on information theory, we propose a novel CFL which seeks to maximize normalized mutual information of the targets and the decision outputs of classifiers. Using the strategy, we can deal with binary/multi-class classifications with/without abstaining. Significant features are observed from the new strategy. While the degree of class imbalance is changing, the proposed strategy is able to balance the errors and rejects accordingly and automatically. Another advantage of the strategy is its ability of deriving optimal rejection thresholds for abstaining classifications and the “equivalent” costs in binary classifications. The connection between rejection thresholds and ROC curve is explored. Empirical investigation is made on several benchmark data sets in comparison with other existing approaches. The classification results demonstrate a promising perspective of the strategy in machine learning.

Index Terms—Classification, class imbalance, cost-free learning, cost-sensitive learning, abstaining, mutual information, ROC.

1 INTRODUCTION

IMBALANCED data sets [1], [2] arise frequently in a variety of real-world applications, such as medicine, biology, finance, and computer vision. Generally, users focus more on the *minority* class and consider the cost of misclassifying a minority class to be more expensive. Unfortunately, most *conventional classification* algorithms assume that the class distributions are balanced or the misclassification costs are equal. They seek to maximize the overall accuracy which yet cannot distinguish the error types. Therefore, they may neglect the significance of the minority class and tend toward the majority class. Learning in the class imbalance is thus of high importance in data mining and machine learning.

From the background of this problem, various methods are developed within a category called *cost-sensitive learning* (CSL), such as costs to test [3], to relabel training instances [4], to sample [5], to weight instances [6], and to find a decision threshold [7], [8]. These methods use unequal costs to make a bias toward the minority class. Generally, when the costs are not given, these methods can not work properly. A comprehensive review of learning in the class imbalance problem is provided by He and Garcia [9].

When there exist some uncertainties in the decision, it may be better to apply *abstaining classification* [10] to

reduce the chance of a potential misclassification. Significant benefits have been obtained from abstaining classification, particularly in very critical applications [11], [12]. The optimal rejection thresholds could be found through minimizing a loss function in a cost-sensitive setting [13], [14], [15]. The possibility of designing loss functions for classifiers with a reject option is also explored [16]. In the context of abstaining classifications, the existing CSL approaches require the cost terms associated to the rejects. However, one often fails to provide such information. Up to now, there seems no proper guideline to give the information in terms of the skew ratio. Obviously, a reject option adds another degree of complexity in classifications over the non-abstaining approaches. For advancing the technology and being compatible with human intelligence, we consider the abstaining strategy will become a common option for most learning machines in future.

In the class imbalance problem, CSL is an important research direction. Based on the definition in [17], we extend it below by including the situation of abstaining.

Definition 1. Cost-Sensitive Learning (CSL) is a type of learning that takes the misclassification costs and/or rejection costs into consideration. The goal of this type of learning is to minimize the total cost.

CSL generally requires modelers or users to specify cost terms for reaching the goal. However, this work addresses one open issue which is mostly overlooked:

“How to conduct a learning in the class imbalance problem when costs are unknown for errors and rejects?”

In fact, the issue is not unusual in real-world applications. Therefore, we propose another category of learning

- X. Zhang is with the National Laboratory of Pattern Recognition, Institute of Automation Chinese Academy of Sciences, Beijing 100190, P.R. China. E-mail: xwzhang@nlpr.ia.ac.cn.
- B.-G. Hu is with the National Laboratory of Pattern Recognition, Institute of Automation Chinese Academy of Sciences, 95 ZhongGuanCun East Road, Beijing 100190, P.R. China. E-mail: hubg@nlpr.ia.ac.cn.

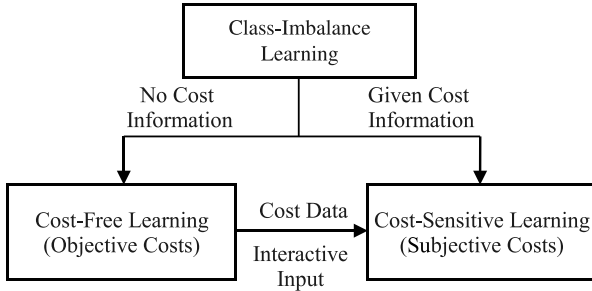


Fig. 1. Cost-Free Learning and Cost-Sensitive Learning.

below for distinguishing the differences between the present work and the existing studies in CSL.

Definition 2. Cost-Free Learning (CFL) is a type of learning that does not require the cost terms associated with the misclassifications and/or rejects as the inputs. The goal of this type of learning is to get optimal classification results without using any cost information.

It is understandable that CFL may face a bigger challenge which is shown by the fact that most existing approaches may fail to present reasonable solutions to the open issue. This work attempts to provide an applicable learning strategy in CFL.

We extend Hu’s [18] study on mutual information classifiers. While Hu presents the theoretical formulas, no learning approaches and results are shown for the real-world data sets. Hence, this work focuses on learning and presents main contributions as follows.

- We propose a CFL strategy in the class imbalance problem. Using *normalized mutual information* (NI) as the learning target, we conduct the learning from cost-insensitive classifiers. Therefore, we are able to adopt conventional classifiers for simple and direct implementations. The most advantage of this strategy is its unique feature in classification scenarios where one has no knowledge of costs.
- We study the relations between the strategy and some existing approaches. First, we derive the “*equivalent*” costs and the rejection thresholds for binary classifications by using the strategy. The costs, being “*objective*” for the reason of purely determined by the distributions of the given data sets, can be a useful reference for “*subjective*” cost specifications in CSL (Fig. 1). Second, we present graphical interpretations of ROC curve plots for both non-abstaining and abstaining classifiers. From the plots, the intrinsic differences between the strategy and other existing approaches are explained in the cases when one class becomes extremely rare.
- We conduct empirical studies on binary class and multi-class problems. Specific investigation is made on abstaining classifications, and we obtain several results from the benchmark data sets which have not been reported before in literature. The results

confirm the advantages of the strategy and show the promising perspective of CFL in imbalanced data sets.

1.1 Related Work

When costs are unequal and unknown, Maloof [20] uses ROC curve to show the performance of binary classifications under different cost settings. The study can be viewed as comparing classifiers rather than finding an optimal operating point. Cost curve [21], [14] can be used to visualize optimal expected costs over a range of cost settings, but it does not suit multi-class problem. Zadrozny and Elkan [22] apply least-squares multiple linear regression to estimate the costs. The method requires cost information of the training sets to be known. Cross validation [23] is proposed to choose from a limited set of cost values, and the final decisions are made by users.

There exists some CFL approaches in the class imbalance problem. Various sampling strategies [24], [25], [26] try to modify the imbalanced class distributions. Active learning [27] is also investigated to select desired instances and the feature selection techniques [28], [29] are applied to combat the class imbalance problem for high-dimensional data sets. Besides, ensemble learning methods [30], [31] are used to improve the generalization of predicting the minority class. The recognition-based methods [32], [33] that train on a single class are proposed as alternatives to the discrimination-based methods to avoid the influence of imbalanced distributions. Sun et al. [34] can get costs for multi-class data sets through maximizing the *geometric mean* (G-mean) or *F-measure* which has the ability of balancing the performance of each class. However, all CFL methods above do not take abstaining into consideration and may fail to process the abstaining classifications.

In regards to abstaining classification, some strategies have been proposed for defining optimal reject rules. Pietraszek [35] proposes a bounded-abstention model with ROC analysis, and Fumera et al. [36] seek to maximize accuracy while keeping the reject rate below a given value. However, the bound information and the targeted reject rate are required to be specified respectively. When there is no prior knowledge of these settings, it is hard to determine the values. Li and Sethi [37] restrict the maximum error rate of each class, but the rates may conflict when they are arbitrarily given.

1.2 Paper Organization

The remainder of this paper is organized as follows: In Section 2, a brief review of NI is provided. We present our CFL strategy in Section 3. Section 4 analyzes the relations between the optimal parameters and the cost terms, and presents the graphical interpretations of ROC curve plots. The experimental results are presented in Section 5. Finally, we conclude this work in Section 6.

TABLE 1
Confusion Matrix C in m -Class Abstaining Classification

T	Y				
	1	2	...	m	$m+1$
1	c_{11}	c_{12}	...	c_{1m}	$c_{1(m+1)}$
2	c_{21}	c_{22}	...	c_{2m}	$c_{2(m+1)}$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
m	c_{m1}	c_{m2}	...	c_{mm}	$c_{m(m+1)}$

2 REVIEW: NORMALIZED MUTUAL INFORMATION

Normalized mutual information (NI) has been used as an evaluation criterion to measure the degree of dependence between the targets T and the decision outputs Y , and it is denoted as

$$NI(T, Y) = \frac{I(T, Y)}{H(T)},$$

where $I(T, Y)$ is the mutual information of two random variables T and Y , $H(T)$ is the Shannon's entropy of T . Note that $NI(T, Y)$ is in the range $[0, 1]$.

Suppose an m -class abstaining classification, with each class denoted as $1, 2, \dots, m$, and the rejected class denoted as $m+1$. The value of the target variable T ranges from 1 to m , while the decision output variable Y ranges from 1 to $m+1$. Then we have

$$I(T, Y) = \sum_{i=1}^m \sum_{j=1}^{m+1} P(T=i, Y=j) \log_2 \frac{P(T=i, Y=j)}{P(T=i)P(Y=j)},$$

$$H(T) = - \sum_{i=1}^m P(T=i) \log_2 P(T=i).$$

In general, as the exact probability distribution functions of T and Y are hard to derive, Hu et al. [38] apply empirical estimations to compute NI based on the confusion matrix. Table 1 illustrates an augmented confusion matrix C in an m -class abstaining classification by adding the last column as a rejected class $m+1$. The rows correspond to the states of the targets T , and the columns correspond to the states of the decision outputs Y . c_{ij} represents the number of the instances that belong to the i -th class classified as the j -th class, $i = 1, 2, \dots, m$, $j = 1, 2, \dots, m+1$. To avoid unchanged value of NI if rejections are made within only one class, the formula of NI is proposed as [38]

$$NI(T, Y) = \frac{\sum_{i=1}^m \sum_{j=1}^m P_e(T=i, Y=j) \frac{P_e(T=i, Y=j)}{P_e(T=i)P_e(Y=j)}}{- \sum_{i=1}^m P_e(T=i) \log_2 P_e(T=i)}$$

$$= - \frac{\sum_{i=1}^m \sum_{j=1}^m c_{ij} \log_2 \left(\frac{c_{ij}}{C_i \sum_{i=1}^m (\frac{c_{ij}}{n})} \right)}{\sum_{i=1}^m C_i \log_2 (\frac{C_i}{n})}, \quad (1)$$

where Y is counted from 1 to m rather than to $m+1$. The subscript “ e ” is given for denoting empirical terms, $C_i = \sum_{j=1}^{m+1} c_{ij}$ is the total number of instances in the i -th class, $i = 1, 2, \dots, m$, and $n = \sum_{i=1}^m \sum_{j=1}^{m+1} c_{ij}$ is the total number in the confusion matrix. In *non-abstaining classification*, i.e. classification without rejection, Y ranges from 1 to m and (1) is actually the formula of the original NI. Then (1) is applicable for both non-abstaining and abstaining classifications in the present work.

Principe et al. [40] present a schematic diagram of *information theory learning* (ITL) and they mention that maximizing mutual information as the target function makes the decision outputs correlate with the targets as much as possible. Recently, a study [18] confirms that ITL opens a new perspective for classifier design. MacKay [39] recommends mutual information for its single rankable value which makes more sense than error rate. Hu et al. [38] study theoretically for the first time on both error types and reject types in binary classifications. They consider information-theoretic measures most promising in providing “objectivity” to classification evaluations in class imbalance problems. The above viewpoints of mutual information motivate our following NI-based strategy for CFL in the class imbalance problem.

3 NI-BASED CLASSIFICATION

In this work, we distinguish two types of classifications, namely, “*non-abstaining classification*” for no rejection and “*abstaining classification*” for rejection. From the phenomenon that different error types and reject types produce different effects on NI, one can derive a conclusion that NI considers the costs to be unequal, unlike accuracy. In fact, the cost information is hiding in NI, and we take advantage of its bias toward the minority class. The bias can be changed through **moving the decision thresholds**, and the value of NI is changed accordingly. We focus our study on the probabilistic classifiers in the present work, although it can also be applied to non-probabilistic classifiers [19].

Let $\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]^T$ denote a data matrix with n instances to be classified, $\mathbf{x}_l \in \mathbb{R}^d$ is the input feature vector, $l = 1, 2, \dots, n$. The target vector is denoted as $\mathbf{t} = [t_1, t_2, \dots, t_n]^T$, $t_l \in T = \{1, 2, \dots, m\}$. The decision output vector is denoted as $\mathbf{y} = [y_1, y_2, \dots, y_n]^T$, $y_l \in Y = \{1, 2, \dots, m\}$ for non-abstaining classification while $y_l \in Y = \{1, 2, \dots, m+1\}$ for abstaining classification. Then for both non-abstaining and abstaining classifications, we have a generalized formula with NI being a function of the data set and the decision thresholds:

$$NI = NI(\mathbf{t}, \mathbf{y} = f(\varphi(\mathbf{x}), \boldsymbol{\tau})),$$

$$y_l = \begin{cases} \arg \max_i \left(\frac{\varphi_i(\mathbf{x}_l)}{\tau_i} \right) & \text{if } \max_i \left(\frac{\varphi_i(\mathbf{x}_l)}{\tau_i} \right) \geq 1, \\ m+1 & \text{otherwise,} \end{cases} \quad (2)$$

$$0 < \tau_i \leq 1, \quad i = 1, 2, \dots, m, \quad l = 1, 2, \dots, n,$$

where $\varphi(\mathbf{x}) \in \mathbb{R}^{n \times m}$ denotes the real-value output matrix of a probabilistic classifier for n instances, $\varphi_i(\mathbf{x}_l)$

is the probabilistic output of class i for $\mathbf{x}_l$, $\sum_{i=1}^m \varphi_i(\mathbf{x}_l) = 1$ and $0 \leq \varphi_i(\mathbf{x}_l) \leq 1$. $\boldsymbol{\tau} = [\tau_1, \tau_2, \dots, \tau_m]^T \in \mathbb{R}^m$ is the vector parameter of the decision thresholds. The decision rule of y_l is proposed in this form to avoid classifying an instance $\mathbf{x}_l$ into more than one class.

3.1 Non-Abstaining Classification

In non-abstaining classification, the first condition for deriving y_l in (2) should only be satisfied, i.e.

$$y_l = \arg \max_i \left(\frac{\varphi_i(\mathbf{x}_l)}{\tau_i} \right), 0 < \tau_i \leq 1, \\ i = 1, 2, \dots, m, l = 1, 2, \dots, n.$$

Let $\phi_i(\mathbf{x}_l) = \alpha_i \varphi_i(\mathbf{x}_l)$, α_i is denoted as the weight parameter for $\varphi_i(\mathbf{x}_l)$, $\alpha_i = \frac{\tau_m}{\tau_i}$ and $\alpha_m = 1$. Then we have the following:

$$\phi_i(\mathbf{x}_l) = \alpha_i \varphi_i(\mathbf{x}_l) \\ = \tau_m \frac{\varphi_i(\mathbf{x}_l)}{\tau_i}.$$

It is obvious that $\arg \max_i \phi_i(\mathbf{x}_l) = \arg \max_i \left(\frac{\varphi_i(\mathbf{x}_l)}{\tau_i} \right)$, and the optimal decision for y_l remains the same. The effect of assigning weights to the probabilistic outputs is the same as setting decision thresholds. Therefore, we denote $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \dots, \alpha_m]^T \in \mathbb{R}^m$ as the weight parameter vector, and the class assignment rule for $y_l = f(\boldsymbol{\varphi}(\mathbf{x}_l), \boldsymbol{\alpha})$ is based on the highest weighted probabilistic outputs. For non-abstaining classification, we propose

$$\begin{aligned} & \text{maximize } NI(\mathbf{t}, \mathbf{y} = f(\boldsymbol{\varphi}(\mathbf{x}), \boldsymbol{\alpha})), \\ & \text{subject to} \\ & y_l = \arg \max_i \alpha_i \varphi_i(\mathbf{x}_l), \\ & \alpha_i > 0, i = 1, 2, \dots, m, l = 1, 2, \dots, n. \end{aligned} \quad (3)$$

In order to maximize NI, the optimal weight parameter $\boldsymbol{\alpha}^*$ should be

$$\boldsymbol{\alpha}^* = \arg \max_{\boldsymbol{\alpha}} NI(\mathbf{t}, \mathbf{y} = f(\boldsymbol{\varphi}(\mathbf{x}), \boldsymbol{\alpha})). \quad (4)$$

3.2 Abstaining Classification

We denote $\mathbf{T}_r = [T_{r1}, T_{r2}, \dots, T_{rm}]^T \in \mathbb{R}^m$ as the rejection threshold vector in dealing with abstaining classification. Let $1 - T_{ri} = \tau_i$, T_{ri} is in the range $[0, 1]$, $i = 1, 2, \dots, m$. The decision output for $y_l = f(\boldsymbol{\varphi}(\mathbf{x}_l), \mathbf{T}_r)$ lies within $m + 1$ classes. Then we propose

$$\begin{aligned} & \text{maximize } NI(\mathbf{t}, \mathbf{y} = f(\boldsymbol{\varphi}(\mathbf{x}), \mathbf{T}_r)), \\ & \text{subject to} \\ & y_l = \begin{cases} \arg \max_i \left(\frac{\varphi_i(\mathbf{x}_l)}{1 - T_{ri}} \right) & \text{if } \max_i \left(\frac{\varphi_i(\mathbf{x}_l)}{1 - T_{ri}} \right) \geq 1, \\ m + 1 & \text{otherwise,} \end{cases} \quad (5) \\ & 0 \leq T_{ri} < 1, 0 \leq \sum_{i=1}^m T_{ri} < m - 1, \\ & i = 1, 2, \dots, m, l = 1, 2, \dots, n. \end{aligned}$$

Note that $m - 1$ is the loose upper bound for the summation $\sum_{i=1}^m T_{ri}$. Assume a situation that all instances satisfy the first condition in (5), and $\frac{\varphi_i(\mathbf{x}_l)}{1 - T_{ri}} \geq 1$ for all probabilistic outputs, i.e. $\forall i, l, \varphi_i(\mathbf{x}_l) \geq 1 - T_{ri}$. Then we get the following:

$$\begin{aligned} \sum_{i=1}^m \varphi_i(\mathbf{x}_l) & \geq \sum_{i=1}^m (1 - T_{ri}), \\ \sum_{i=1}^m T_{ri} & \geq m - 1. \end{aligned}$$

If $\sum_{i=1}^m T_{ri}$ falls in this interval, the condition of rejection would never be satisfied and the proposal of abstaining classification is ineffective. Reversely, this extreme situation would not happen if $\sum_{i=1}^m T_{ri} < m - 1$.

In order to maximize NI, the optimal rejection threshold vector $\mathbf{T}_r^*$ should be

$$\mathbf{T}_r^* = \arg \max_{\mathbf{T}_r} NI(\mathbf{t}, \mathbf{y} = f(\boldsymbol{\varphi}(\mathbf{x}), \mathbf{T}_r)). \quad (6)$$

3.3 Optimization Algorithm

The present framework is proposed based on the confusion matrix from which we compute NI, but it is not differentiable. We apply a general optimization algorithm called “*Powell Algorithm*” which is a direct method for nonlinear optimization without calculating the derivatives [41]. It is also widely used in image registration to find optimal registration parameters.

Algorithm 1 Learning algorithm

Input: Probabilistic outputs $\boldsymbol{\varphi}(\mathbf{x})$, target labels $\mathbf{t}$, $\mathcal{D}$ as the degree of freedom in $\boldsymbol{\tau}$.

Output: $\boldsymbol{\tau}^*$

1: Initialize $\boldsymbol{\tau}_1$ as a random vector in the range of $\boldsymbol{\tau}$, $\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_{\mathcal{D}}$ as linear independent vectors, number of iterations $W = 0$, $\varepsilon \geq 0$.

2: **Iterative Search Phase:**

3: **repeat**

4: $W = W + 1$. Let $\boldsymbol{\tau}_W^{(1)} = \boldsymbol{\tau}_W$.

5: **for** each direction $\mathbf{d}_i$, $i = 1$ **to** $\mathcal{D}$ **do**

6: $\bar{\eta}^{(i)} = \arg \max_{\eta \in \mathbb{R}} NI(\mathbf{t}, \mathbf{y} = f(\boldsymbol{\varphi}(\mathbf{x}), \boldsymbol{\tau}_W^{(i)} + \eta \mathbf{d}_i));$

7: Update $\boldsymbol{\tau}_W$ in the current direction: $\boldsymbol{\tau}_W^{(i+1)} = \boldsymbol{\tau}_W^{(i)} + \bar{\eta}^{(i)} \mathbf{d}_i;$

8: **end for**

9: Update the directions: $\mathbf{d}_i = \mathbf{d}_{i+1}, i = 1, 2, \dots, \mathcal{D} - 1;$

$\mathbf{d}_{\mathcal{D}} = \boldsymbol{\tau}_W^{(\mathcal{D}+1)} - \boldsymbol{\tau}_W;$

10: $\eta_W^* = \arg \max_{\eta \in \mathbb{R}} NI(\mathbf{t}, \mathbf{y} = f(\boldsymbol{\varphi}(\mathbf{x}), \boldsymbol{\tau}_W + \eta \mathbf{d}_{\mathcal{D}}));$

11: Update $\boldsymbol{\tau}$ after the current iteration: $\boldsymbol{\tau}_{W+1} = \boldsymbol{\tau}_W + \eta_W^* \mathbf{d}_{\mathcal{D}};$

12: **until** $\|\boldsymbol{\tau}_{W+1} - \boldsymbol{\tau}_W\|_2 \leq \varepsilon$

13: Return $\boldsymbol{\tau}^* = \boldsymbol{\tau}_{W+1}$.

The algorithm is given in Algorithm 1, which we apply to find $\boldsymbol{\tau}^*$ for demonstration. We can also apply it to both $\boldsymbol{\alpha}$ and $\mathbf{T}_r$. For Step 6 and Step 10, we use *bracketing method* to find three starting points and use *Brent's Method* to realize one-dimensional optimization. W iterations of the basic procedure lead to $W(\mathcal{D} + 1)$ one-dimensional optimizations. One disadvantage of this algorithm is that it may find a local extrema. Hence, we randomly choose the starting points several times and then pick the best one. In non-abstaining binary

classification, $\mathcal{D} = 1$, so we just work from Step 4 to Step 7 once and assign the value of $\tau_W^{(2)}$ to τ^* .

4 RELATIONS IN BINARY CLASSIFICATION

The previous section completes the essence of the present framework. It can be regarded as a generic way to make the conventional learning algorithms information-theoretic based.

The optimal parameters reflect the degree of bias implied by NI, and may reveal the cost information to some extent. In this section, we focus on binary classification and analyze the relations between the optimal parameters and the cost terms. Moreover, we discover some graphical interpretations of performance measures on ROC curve, which allows the users to adjust the parameters more conveniently using ROC curve.

4.1 Normalized Cost Matrix

Friedel et al. [14] derive normalized cost matrix based on the *overall risk* which is written as

$$Risk = \sum_{i,j} \lambda_{ij} p(j|i) p(i), \quad (7)$$

where λ_{ij} is the original cost in the common cost matrix that assigns an instance of class i to class j , $p(j|i)$ is the true probability in such situation, and $p(i)$ is the true prior probability of class i . The *conditional risk* of assigning an instance x_l to class j is

$$Risk(j|x_l) = \sum_{i=1}^m \lambda_{ij} p(i|x_l), \quad (8)$$

where $p(i|x_l)$ is the true posterior probability of class i given x_l . By applying the way of transforming costs [14], we find that the normalization way for the overall risk is also applicable for the conditional risk.

In binary classification, we refer to class 1 and class 2 as *negative class* (N) and *positive class* (P), respectively. We denote λ_{FN} , λ_{FP} , λ_{TN} , λ_{TP} , λ_{RN} and λ_{RP} to be the costs of *false negative*, *false positive*, *true negative*, *true positive*, *reject negative*, and *reject positive*, respectively. Therefore, the normalized cost matrix for non-abstaining binary classification can be denoted as

$$\bar{\lambda}_{no_rej} = \begin{bmatrix} \bar{\lambda}_{TN} & \bar{\lambda}_{FP} \\ \bar{\lambda}_{FN} & \bar{\lambda}_{TP} \end{bmatrix} = \begin{bmatrix} 0 & \bar{\lambda}_{FP} \\ 1 & 0 \end{bmatrix} \quad (9)$$

with $\beta = \lambda_{FN} - \lambda_{TP}$, then $\bar{\lambda}_{TN} = \frac{\lambda_{TN} - \lambda_{TN}}{\beta} = 0$, $\bar{\lambda}_{FP} = \frac{\lambda_{FP} - \lambda_{TN}}{\beta}$, $\bar{\lambda}_{FN} = \frac{\lambda_{FN} - \lambda_{TP}}{\beta} = 1$, $\bar{\lambda}_{TP} = \frac{\lambda_{TP} - \lambda_{TP}}{\beta} = 0$.

Similarly, the normalized cost matrix for abstaining binary classification can be denoted as

$$\bar{\lambda}_{rej} = \begin{bmatrix} \bar{\lambda}_{TN} & \bar{\lambda}_{FP} & \bar{\lambda}_{RN} \\ \bar{\lambda}_{FN} & \bar{\lambda}_{TP} & \bar{\lambda}_{RP} \end{bmatrix} = \begin{bmatrix} 0 & \bar{\lambda}_{FP} & \bar{\lambda}_{RN} \\ 1 & 0 & \bar{\lambda}_{RP} \end{bmatrix} \quad (10)$$

with $\bar{\lambda}_{TN} = 0$, $\bar{\lambda}_{FP} = \frac{\lambda_{FP} - \lambda_{TN}}{\beta}$, $\bar{\lambda}_{RN} = \frac{\lambda_{RN} - \lambda_{TN}}{\beta}$, $\bar{\lambda}_{FN} = 1$, $\bar{\lambda}_{TP} = 0$, $\bar{\lambda}_{RP} = \frac{\lambda_{RP} - \lambda_{TP}}{\beta}$, $\beta = \lambda_{FN} - \lambda_{TP}$. The first two columns contain the misclassification costs, while the last column indicates the rejection costs.

It is reasonable to assume that the values of the original correct classification costs and misclassification costs in the common cost matrix are not affected by introducing a reject option. Therefore, what is noteworthy is that $\bar{\lambda}_{FP}$ in (9) is consistent with that in (10).

4.2 Optimal Weight and Misclassification Cost

In non-abstaining binary classification, it is feasible to set the decision thresholds as $\tau = [1 - \tau_P, \tau_P]^T$, which has one degree of freedom.

The relation between the decision thresholds and the costs has been derived by Elkan [7] through minimizing the conditional risk. Considering the normalized cost matrix in (9), the decision threshold τ_P^* of the positive class for making optimal decision can be represented as

$$\tau_P^* = \frac{\bar{\lambda}_{FP}}{1 + \bar{\lambda}_{FP}}, \quad (11)$$

with $\bar{\lambda}_{FP}$ be the variable. It is required that the value of $\bar{\lambda}_{FP}$ be given and be reasonable. Otherwise, τ_P^* can not be derived or not be proper.

In our present work, the optimal weight vector is $\alpha^* = [\alpha_N^*, 1]^T$. We apply it in the decision rule of maximum weighted posterior probability. Then the optimal prediction is the positive class if and only if $\alpha_N^* p(N|x_l) \leq p(P|x_l)$. Hence, the decision threshold τ_P^{**} of the positive class for making optimal decision is

$$\tau_P^{**} = \frac{\alpha_N^*}{1 + \alpha_N^*}. \quad (12)$$

Suppose that the minimum conditional risk rule shares the same decision thresholds with the maximum weighted posterior probability rule, then (11) and (12) should be equal. And we give the following definition:

Definition 3. Given the optimal weight α_N^* , the “equivalent” misclassification cost is defined as

$$\bar{\lambda}_{FP} = \alpha_N^*. \quad (13)$$

In general, it is assumed that $\bar{\lambda}_{FP} < \bar{\lambda}_{FN}$, i.e. $\bar{\lambda}_{FP} < 1$. In this case, it is required that $\alpha_N^* < 1$.

4.3 Optimal Rejection Thresholds and Costs

In abstaining binary classification, the relations between the rejection thresholds and the costs can be presented in a form of explicit formulae [18]. With the optimal rejection threshold vector $\mathbf{T}_r^* = [T_{rN}^*, T_{rP}^*]^T$ and the normalized cost matrix in (10), these relations are

$$T_{rN}^* = \frac{\bar{\lambda}_{RN}}{1 + \bar{\lambda}_{RN} - \bar{\lambda}_{RP}},$$

$$T_{rP}^* = \frac{\bar{\lambda}_{RP}}{\bar{\lambda}_{FP} - \bar{\lambda}_{RN} + \bar{\lambda}_{RP}},$$

which imply a parameter redundancy. In addition, the value of $\bar{\lambda}_{FP}$ derived from (13) can be utilized as a prior knowledge under the assumption of cost consistency.

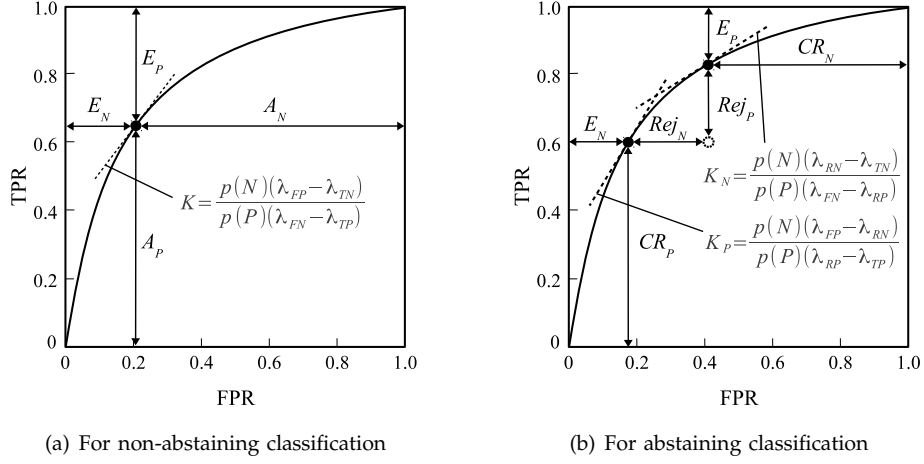


Fig. 2. Graphical interpretations of ROC curves. (a) For non-abstaining classification. (b) For abstaining classification.

Definition 4. Given the “equivalent” misclassification cost $\bar{\lambda}_{FP} = \alpha_N^*$, the “equivalent” rejection costs are defined as

$$\begin{aligned}\bar{\lambda}_{RN} &= \frac{T_{rN}^*(1 - T_{rP}^*) - T_{rN}^* T_{rP}^* \bar{\lambda}_{FP}}{1 - T_{rN}^* - T_{rP}^*}, \\ \bar{\lambda}_{RP} &= \frac{-T_{rN}^* T_{rP}^* + (1 - T_{rN}^*) T_{rP}^* \bar{\lambda}_{FP}}{1 - T_{rN}^* - T_{rP}^*}.\end{aligned}\quad (14)$$

Based on [18], one can have the relations $\bar{\lambda}_{TN} < \bar{\lambda}_{RN} < \bar{\lambda}_{FP}$ and $\bar{\lambda}_{TP} < \bar{\lambda}_{RP} < \bar{\lambda}_{FN}$. Then we can obtain the following properties from (14):

- P1. If $0 < \bar{\lambda}_{RN} < \bar{\lambda}_{FP}$, we have $0 < T_{rN}^* < \frac{\alpha_N^*}{1 + \alpha_N^*}$ and $T_{rP}^* < \frac{1}{1 + \alpha_N^*}$;
- P2. If $0 < \bar{\lambda}_{RP} < 1$, we have $T_{rN}^* < \frac{\alpha_N^*}{1 + \alpha_N^*}$ and $0 < T_{rP}^* < \frac{1}{1 + \alpha_N^*}$;
- P3. If $0 < T_{rN}^* < \frac{\alpha_N^*}{1 + \alpha_N^*}$ and $0 < T_{rP}^* < \frac{1}{1 + \alpha_N^*}$, then $0 < \bar{\lambda}_{RN} < \bar{\lambda}_{FP}$ and $0 < \bar{\lambda}_{RP} < 1$.

4.4 Graphical Interpretations of ROC Curve Plots with/without Abstaining

In binary classification, an ROC curve plot presents complete information about the performance of each class [42], so that an *overall performance measure* [43], such as AUC, can be formed. This is a preferred feature in processing class imbalance problems [44]. Furthermore, an ROC curve can also provide the graphical interpretations for non-abstaining and abstaining classifications in Fig. 2, where TPR and FPR are *true positive rate* and *false positive rate*. We denote A , CR , E and Rej to be *accuracy*, *correct recognition rate*, *error rate*, and *reject rate*, respectively. C_N and C_P are the total numbers of the negatives and positives, respectively. C_{FN} , C_{FP} , C_{TN} , C_{TP} , C_{RN} and C_{RP} are the numbers of the *false negatives*, *false positives*, *true negatives*, *true positives*, *reject negatives*, and *reject positives*, respectively. Their relations are shown as follows:

Non-abstaining:

$$\begin{aligned}A_N + E_N &= 1, \text{ and } A_P + E_P = 1, \\ A_N &= \frac{C_{TN}}{C_N}, E_N = \frac{C_{FP}}{C_N}, A_P = \frac{C_{TP}}{C_P}, E_P = \frac{C_{FN}}{C_P};\end{aligned}\quad (15a)$$

Abstaining:

$$\begin{aligned}CR_N + E_N + Rej_N &= 1, \text{ and } CR_P + E_P + Rej_P = 1, \\ CR_N &= \frac{C_{TN}}{C_N}, E_N = \frac{C_{FP}}{C_N}, Rej_N = \frac{C_{RN}}{C_N}, \\ CR_P &= \frac{C_{TP}}{C_P}, E_P = \frac{C_{FN}}{C_P}, Rej_P = \frac{C_{RP}}{C_P}.\end{aligned}\quad (15b)$$

Several observations are summarized below for understanding the features of ROC plots. To begin with, we discuss an ROC curve in a non-abstaining classification, as shown in Fig. 2a. For a theoretical ROC curve which is concave, the decision is made by K , the *slope of ROC curve*, in the form of [45]:

$$K = \frac{p(N) \lambda_{FP} - \lambda_{TN}}{p(P) \lambda_{FN} - \lambda_{TP}} = \frac{p(N)}{p(P)} \bar{\lambda}_{FP}, \quad (16)$$

which is also equivalent to the likelihood ratio [46]:

$$L = \frac{p(x|P)}{p(x|N)} = \frac{p(N) \lambda_{FP} - \lambda_{TN}}{p(P) \lambda_{FN} - \lambda_{TP}} = \frac{p(N)}{p(P)} \bar{\lambda}_{FP}. \quad (17)$$

From (16), one can observe that:

$$\text{if } p(P) \rightarrow 0, \text{ then } K \rightarrow \infty, \quad (18a)$$

$$\text{and } E_P = 1, A_P = 0, E_N = 0, A_N = 1, \quad (18b)$$

for general cost terms. (18a) indicates that the tangent point on the ROC curve will be located at the origin in Fig. 2a, and (18b) demonstrates a graphical interpretation why conventional classifiers fail to process minority class (herein the positive class) properly. However, the situation in (18) can never appear from using the present strategy, because it will result in a zero value of mutual information [39], [43].

Different with the non-abstaining classification, Fig. 2b shows the abstaining classification graphically on an ROC curve. Two *abstaining slopes*, K_N and K_P , are generally given in the forms of [15]:

$$\begin{aligned}K_N &= \frac{p(N) \lambda_{RN} - \lambda_{TN}}{p(P) \lambda_{FN} - \lambda_{RP}} = \frac{p(N)}{p(P)} \frac{\bar{\lambda}_{RN}}{1 - \bar{\lambda}_{RP}}, \\ K_P &= \frac{p(N) \lambda_{FP} - \lambda_{RN}}{p(P) \lambda_{RP} - \lambda_{TP}} = \frac{p(N)}{p(P)} \frac{\bar{\lambda}_{FP} - \bar{\lambda}_{RN}}{\bar{\lambda}_{RP}}.\end{aligned}\quad (19)$$

TABLE 2
Description of the Data Sets

Data Set	#Inst	#Attr	#C	Class Distribution
Isr	11,180	7	2	10,920/260(=42.00)
Nursery(very_recom)	12,960	9	2	12,632/328(=38.51)
Letter(A)	20,000	17	2	19,211/789(=24.35)
Rooftop	17,829	10	2	17,048/781(=21.83)
Pendigits(5)	10,992	17	2	9,937/1,055(=9.42)
Optdigits(8)	5,620	65	2	5,066/554(=9.14)
Vehicle(opel)	846	19	2	634/212(=2.99)
Yeast(NUC)	1,484	10	2	1,055/429(=2.46)
Phoneme	5,404	6	2	3,818/1,586(=2.41)
German Credit	1,000	25	2	700/300(=2.33)
Diabetes	768	9	2	500/268(=1.87)
Gamma	19,020	11	2	12,332/6,688(=1.84)
Cardiotocography	2,126	22	3	1,655/295/176
Thyroid	7,200	22	3	6,666/368/166
Car	1,728	7	4	1,210/384/65/69
Pageblock	5,473	11	5	4,913/329/28/88/115

(#Inst: number of instances, #Attr: number of attributes, #C: number of classes)

Whenever $K_N \neq K_P$, one can observe the non-zero results of rejection rates. (19) confirms the finding in [18] that at most two independent parameters will determine the rejection range in binary classifications. Sometimes, one can still apply a single independent parameter, such as $K_P = 2K_N$, for abstaining decisions.

There exist relations between rejection thresholds in the *posterior curve plot* [18] and abstaining slopes in the *ROC curve plot*. Their relations and the associated constraint are derived from [18]:

$$K_N = \frac{p(N)}{p(P)} \frac{T_{rN}}{1 - T_{rN}}, \quad K_P = \frac{p(N)}{p(P)} \frac{1 - T_{rP}}{T_{rP}}, \quad K_N < K_P. \quad (20)$$

5 EXPERIMENTS

5.1 Configuration

Table 2 lists twelve binary class and four multi-class data sets with imbalanced class distributions. On *Pageblock*, the maximum ratio between the majority class and the minority class is 175. Most of the data sets are obtained from the UCI Machine Learning Repository¹, *Isr* is from [25], *Rooftop* is from [20], and *Phoneme* is from KEEL Datasets². All of them have continuous attributes and are rescaled to be in the range [0, 1]. We perform 3-fold cross validation and all experiments are repeated ten times to get the average results. In addition, Table 3 lists the procedure of our NI based experiments for each run.

We call our NI based non-abstaining classification and NI based abstaining classification “*NI_no_rej*” and “*NI_rej*” respectively. To illustrate the effectiveness of our strategy, we adopt *kNN* and *Bayes classifier* as the conventional classifiers, and we compare our methods with *SMOTE*, *Cost-sensitive learning*, *Chow’s reject* [10] methods and the G-mean based methods (“*Gmean_no_rej*” and “*Gmean_rej*”) besides two conventional classifications.

In *kNN classifier*, we apply Euclidean distance and use the confidence values [47], [19] as the probabilistic

TABLE 3
The Procedure of Our NI Based Experiments

1. Apply 3-fold cross validation on a data set. $\frac{2}{3}$ data belong to the training set and the remainder belong to the test set.
 - a. Apply 3-fold cross validation on the training set. $\frac{2}{3}$ data belong to the estimation set and the remainder belong to the validation set.
 - i. Apply Algorithm 1 several times to get the best parameter in each cross validation.
 - b. Apply the mean value of 3 best parameters in step a to the training set.
 - c. Predict the test set with the parameter obtained from step b.
2. Obtain the results of 3 test sets.

outputs. The class assignment is decided by the highest confidence. For brevity, we just list the results of 11-NN on all data sets except 5-NN on *Pageblock*. In *Bayes classifier*, we derive the estimated class-conditional density from the Parzen-window estimation with Gaussian kernel [48] and apply Bayes rule to classification. The smooth parameter is chosen as the average value of the distance from one instance to its r th nearest neighborhood ($r=10$ empirically), and the empirical probability of the occurrence of class is chosen as the prior probability. In *SMOTE*, the average results are presented with the amount from 1 to 5, and it performs simultaneously on the minority classes of the multi-class data sets with the same amount. In *Cost-sensitive learning*, we simply assign the inverse of the class distribution ratio to the misclassification cost λ_{ij} for $i \neq j$, and $\lambda_{ii}=0$. We do not consider abstaining for it because the rejection costs would be hard to give. In *Chow’s reject*, we simply assign 0.3 to the rejection thresholds for all classes. In G-mean based methods, we apply our way of parameter settings and optimization to maximize G-mean.

5.2 Evaluation Criterion

In order to show the changes of each class clearly, E_i and Rej_i are applied as the error rate and the reject rate within its i th class respectively. The total error rate (“*E*”) and the total reject rate (“*Rej*”) are also applied. “*A*” is short for the total accuracy. “*G*” is short for G-mean with the formula $G-mean = (\prod_{i=1}^m A_i)^{\frac{1}{m}}$, where A_i represents the accuracy within its i th class. In binary class tasks, we also evaluate F-measure (“*F*” for short).

5.3 Binary Class Tasks

The results on the binary class data sets are shown in Table 4. Both conventional classifiers have high accuracies and low error rates of the negative class, but the error rates of the positive class are high. *SMOTE* is an effective method with low error rate of the positive class. However, it does not have the ability to reject instances. *Cost-sensitive learning* performs well under the current cost settings, but its accuracy is the lowest when the class distribution differs greatly. On *Nursery* and *Letter*, the error rate of the positive class is zero with *Cost-sensitive learning* at the price of high error rate of the negative class. Besides, *Gmean_no_rej* and *NI_no_rej* perform well on balancing the classification of two classes. When a

1. <http://archive.ics.uci.edu/ml/>

2. <http://sci2s.ugr.es/keel/datasets.php>

TABLE 4

Evaluation Results on Binary Class Data Sets, “—”: Not Available, The Best Performance in Each Cell is Bolded

Data set		Method	$E_N(\%)$	$E_P(\%)$	$E(\%)$	$A(\%)$	$Rej_N(\%)$	$Rej_P(\%)$	$Rej(\%)$	$G(\%)$	$F(\%)$	NI
Ism	k N N	kNN classifier	0.27	49.03	1.40	98.60	—	—	—	71.22	62.71	0.3763
		SMOTE	1.13	33.17	1.87	98.13	—	—	—	81.13	62.66	0.4361
		Cost-sensitive	4.66	17.33	4.96	95.04	—	—	—	88.74	43.78	0.4101
		Gmean_no_rej	4.11	17.61	4.42	95.58	—	—	—	88.87	46.57	0.4251
		NI_no_rej	1.13	29.94	1.80	98.20	—	—	—	83.17	64.69	0.4612
		Chow's reject	0.11	37.84	0.99	99.00	0.45	23.93	0.99	70.86	64.31	0.3260
		Gmean_rej	2.72	17.54	3.06	96.88	1.91	4.62	1.97	89.04	54.50	0.4553
		NI_rej	0.82	18.85	1.24	98.71	3.10	15.88	3.40	87.67	71.04	0.4815
	B a y e s	Bayes classifier	0.23	50.28	1.40	98.60	—	—	—	70.34	62.20	0.3707
		SMOTE	0.77	36.37	1.60	98.40	—	—	—	79.27	64.79	0.4353
		Cost-sensitive	12.38	12.77	12.39	87.61	—	—	—	87.40	24.74	0.2960
		Gmean_no_rej	7.50	16.68	7.71	92.29	—	—	—	87.73	35.99	0.3567
		NI_no_rej	1.27	28.15	1.90	98.10	—	—	—	84.18	63.82	0.4664
		Chow's reject	0.08	39.92	1.01	98.98	0.44	19.09	0.87	71.05	65.22	0.3489
		Gmean_rej	1.60	9.37	1.78	96.63	41.53	18.42	40.99	92.61	68.39	0.4761
		NI_rej	1.10	16.08	1.44	98.45	6.82	12.96	6.96	89.74	69.71	0.5011
Nursery	k N N	kNN classifier	0.02	38.75	1.00	99.00	—	—	—	78.23	75.56	0.5270
		SMOTE	1.45	12.09	1.72	98.28	—	—	—	92.83	72.86	0.6415
		Cost-sensitive	5.79	0.61	5.66	94.34	—	—	—	96.76	47.10	0.5614
		Gmean_no_rej	2.83	1.89	2.81	97.19	—	—	—	97.63	64.03	0.6608
		NI_no_rej	0.79	9.39	1.00	99.00	—	—	—	94.80	82.11	0.7154
		Chow's reject	0.02	11.71	0.31	99.68	0.36	56.82	1.79	85.31	83.53	0.3753
		Gmean_rej	0.16	0.73	0.17	99.82	4.86	36.90	5.67	99.38	96.15	0.7450
		NI_rej	0.57	2.10	0.61	99.37	2.53	9.72	2.71	98.53	88.23	0.7686
	B a y e s	Bayes classifier	0.00	100.00	2.52	97.48	—	—	—	0.00	0.00	0.0000
		SMOTE	0.04	85.42	2.20	97.80	—	—	—	28.03	21.64	0.1117
		Cost-sensitive	29.95	0.00	29.19	70.81	—	—	—	83.69	14.78	0.2523
		Gmean_no_rej	7.14	2.56	7.03	92.97	—	—	—	95.10	41.72	0.5002
		NI_no_rej	3.24	12.29	3.47	96.53	—	—	—	92.05	56.90	0.5219
		Chow's reject	0.00	98.84	2.50	97.50	0.01	1.16	0.04	0.00	0.00	0.0000
		Gmean_rej	0.04	0.12	0.04	99.95	16.28	79.58	17.88	99.67	95.98	0.4817
		NI_rej	1.52	0.88	1.50	98.35	8.69	23.32	9.06	98.61	72.71	0.5890
Letter	k N N	kNN classifier	0.02	4.21	0.19	99.81	—	—	—	97.86	97.56	0.9192
		SMOTE	0.17	1.03	0.20	99.80	—	—	—	99.40	97.47	0.9418
		Cost-sensitive	1.18	0.00	1.13	98.87	—	—	—	99.41	87.44	0.8379
		Gmean_no_rej	0.52	0.18	0.51	99.49	—	—	—	99.65	93.99	0.9029
		NI_no_rej	0.13	1.14	0.17	99.83	—	—	—	99.36	97.91	0.9452
		Chow's reject	0.01	1.39	0.06	99.94	0.09	6.62	0.35	99.25	99.15	0.9189
		Gmean_rej	0.01	0.05	0.02	99.98	0.83	8.14	1.12	99.97	99.79	0.9561
		NI_rej	0.06	0.22	0.07	99.93	0.42	2.45	0.50	99.86	99.16	0.9585
	B a y e s	Bayes classifier	0.00	100.00	3.95	96.05	—	—	—	0.00	0.00	0.0000
		SMOTE	0.02	84.76	3.36	96.64	—	—	—	24.65	20.72	0.1229
		Cost-sensitive	14.56	0.00	13.99	86.01	—	—	—	92.43	36.12	0.4318
		Gmean_no_rej	2.14	12.34	2.54	97.46	—	—	—	92.61	73.31	0.5996
		NI_no_rej	0.78	14.87	1.34	98.66	—	—	—	91.90	83.41	0.6687
		Chow's reject	0.00	14.04	0.55	99.44	0.11	43.65	1.83	86.65	85.76	0.4598
		Gmean_rej	0.01	0.03	0.01	99.96	59.64	90.98	60.88	99.76	98.80	0.1784
		NI_rej	0.49	11.18	0.91	99.05	5.12	7.92	5.23	93.52	87.66	0.6780
Rooftop	k N N	kNN classifier	0.59	76.99	3.94	96.06	—	—	—	47.78	33.82	0.1264
		SMOTE	4.20	54.28	6.39	93.61	—	—	—	65.36	38.40	0.1779
		Cost-sensitive	19.12	21.03	19.20	80.80	—	—	—	79.90	26.48	0.1926
		Gmean_no_rej	18.53	21.25	18.65	81.35	—	—	—	80.07	27.07	0.1964
		NI_no_rej	8.46	37.99	9.75	90.25	—	—	—	75.28	35.97	0.2027
		Chow's reject	0.18	63.71	2.97	96.96	1.57	23.56	2.54	40.66	27.23	0.0917
		Gmean_rej	8.30	20.92	8.85	90.08	10.63	17.62	10.94	82.19	38.09	0.2345
		NI_rej	3.36	24.67	4.29	94.97	13.69	28.84	14.35	79.45	48.85	0.2388
	B a y e s	Bayes classifier	0.35	81.68	3.92	96.08	—	—	—	42.64	29.01	0.1063
		SMOTE	1.89	63.31	4.58	95.42	—	—	—	59.19	40.37	0.1744
		Cost-sensitive	17.33	19.74	17.43	82.57	—	—	—	81.45	28.77	0.2170
		Gmean_no_rej	17.53	19.00	17.59	82.41	—	—	—	81.71	28.82	0.2009
		NI_no_rej	10.27	33.14	11.27	88.73	—	—	—	76.96	35.64	0.2176
		Chow's reject	0.06	67.37	3.01	96.92	1.10	25.56	2.17	30.67	17.01	0.0587
		Gmean_rej	1.30	2.25	1.34	95.15	71.17	66.75	70.98	94.25	69.59	0.1895
		NI_rej	3.53	12.02	3.90	94.70	25.92	38.73	26.48	87.57	53.07	0.2679
Pendigits	k N N	kNN classifier	0.16	1.01	0.24	99.76	—	—	—	99.41	98.73	0.9525
		SMOTE	0.28	0.50	0.30	99.70	—	—	—	99.61	98.45	0.9509
		Cost-sensitive	0.55	0.34	0.53	99.47	—	—	—	99.55	97.29	0.9284
		Gmean_no_rej	0.31	0.46	0.33	99.67	—	—	—	99.62	98.32	0.9482
		NI_no_rej	0.24	0.59	0.27	99.73	—	—	—	99.59	98.60	0.9536
		Chow's reject	0.08	0.63	0.14	99.86	0.18	1.10	0.27	99.64	99.28	0.9625
		Gmean_rej	0.08	0.27	0.09	99.90	0.69	3.47	0.96	99.82	99.50	0.9635
		NI_rej	0.10	0.40	0.13	99.87	0.32	1.08	0.40	99.75	99.31	0.9655
	B a y e s	Bayes classifier	0.14	9.01	0.99	99.01	—	—	—	95.32	94.62	0.8245
		SMOTE	1.09	2.64	1.24	98.76	—	—	—	98.11	93.86	0.8536
		Cost-sensitive	4.89	0.15	4.43	95.57	—	—	—	97.45	81.22	0.7200
		Gmean_no_rej	1.11	0.89	1.09	98.91	—	—	—	99.00	94.60	0.8746
		NI_no_rej	0.62	2.27	0.78	99.22	—	—	—	98.55	96.01	0.8843
		Chow's reject	0.02	2.07	0.22	99.78	0.55	22.75	2.68	98.64	98.52	0.7887
		Gmean_rej	0.03	0.06	0.03	99.96	5.80	27.52	7.89	99.94	99.78	0.8414
		NI_rej	0.36	0.52	0.38	99.61	2.58	4.10	2.72	99.54	97.98	0.9061

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Data set	Method		$E_N(\%)$	$E_P(\%)$	$E(\%)$	$A(\%)$	$Rej_N(\%)$	$Rej_P(\%)$	$Rej(\%)$	$G(\%)$	$F(\%)$	NI
Optdigits	kNN	kNN classifier	0.10	4.80	0.56	99.44	—	—	—	97.52	97.08	0.8938
		SMOTE	0.59	1.65	0.69	99.31	—	—	—	98.88	96.57	0.9019
		Cost-sensitive	1.09	1.19	1.10	98.90	—	—	—	98.86	94.65	0.8709
		Gmean_no_rej	0.84	1.34	0.89	99.11	—	—	—	98.91	95.64	0.8868
		NI_no_rej	0.28	2.26	0.48	99.52	—	—	—	98.72	97.59	0.9160
		Chow's reject	0.06	2.35	0.28	99.72	0.21	7.29	0.90	98.70	98.45	0.8898
		Gmean_rej	0.07	0.76	0.14	99.86	1.85	10.06	2.66	99.54	99.25	0.9221
		NI_rej	0.15	1.41	0.27	99.72	0.74	2.04	0.87	99.21	98.58	0.9307
	Bayes	Bayes classifier	0.00	100.00	9.86	90.14	—	—	—	0.00	0.00	0.0000
		SMOTE	1.29	60.59	7.14	92.86	—	—	—	46.68	40.41	0.2741
		Cost-sensitive	52.14	0.00	47.00	53.00	—	—	—	69.18	29.56	0.1855
		Gmean_no_rej	6.19	3.90	5.96	94.04	—	—	—	94.94	76.13	0.6109
		NI_no_rej	4.39	8.26	4.77	95.23	—	—	—	93.62	79.18	0.6134
		Chow's reject	0.00	100.00	9.86	90.14	0.00	0.00	0.00	0.00	0.00	0.0000
		Gmean_rej	0.08	0.18	0.09	99.86	39.84	59.48	41.78	99.70	99.02	0.4653
		NI_rej	2.50	1.53	2.40	97.31	10.44	15.21	10.91	97.69	87.48	0.6626
Vehicle	kNN	kNN classifier	5.81	68.30	21.47	78.53	—	—	—	54.46	42.35	0.0915
		SMOTE	30.69	24.93	29.25	70.75	—	—	—	70.47	55.69	0.1542
		Cost-sensitive	21.33	34.90	24.73	75.27	—	—	—	71.45	56.95	0.1442
		Gmean_no_rej	29.47	24.50	28.23	71.77	—	—	—	72.79	57.28	0.1523
		NI_no_rej	39.57	14.94	33.40	66.60	—	—	—	71.14	56.21	0.1579
		Chow's reject	0.79	36.43	9.72	86.61	19.09	52.44	27.44	47.72	36.05	0.1039
		Gmean_rej	3.97	3.49	3.85	90.71	55.62	74.26	60.29	88.49	74.85	0.1764
		NI_rej	14.67	6.86	12.71	80.69	34.63	42.54	36.61	82.46	68.11	0.1982
	Bayes	Bayes classifier	0.62	93.13	23.80	76.20	—	—	—	23.22	11.94	0.0288
		SMOTE	30.04	41.96	33.03	66.97	—	—	—	57.28	42.86	0.0755
		Cost-sensitive	26.97	40.42	30.33	69.67	—	—	—	65.86	49.54	0.0790
		Gmean_no_rej	36.91	25.71	34.09	65.91	—	—	—	68.24	52.08	0.1034
		NI_no_rej	45.48	17.48	38.46	61.54	—	—	—	66.52	51.85	0.1072
		Chow's reject	0.00	48.42	12.13	82.85	21.86	51.58	29.31	0.00	0.00	0.0236
		Gmean_rej	2.40	1.60	2.20	91.40	73.51	82.10	75.66	84.71	73.28	0.1223
		NI_rej	9.64	8.30	9.30	81.23	50.04	58.16	52.07	79.80	65.67	0.1322
Yeast	kNN	kNN classifier	9.44	58.16	23.52	76.48	—	—	—	61.50	50.64	0.1086
		SMOTE	36.11	25.29	32.98	67.02	—	—	—	67.48	56.39	0.1180
		Cost-sensitive	25.08	32.82	27.32	72.68	—	—	—	70.92	58.70	0.1289
		Gmean_no_rej	30.99	26.62	29.73	70.27	—	—	—	71.02	58.84	0.1289
		NI_no_rej	31.52	26.25	30.00	70.00	—	—	—	70.35	58.72	0.1340
		Chow's reject	3.07	34.17	12.06	83.31	20.68	45.41	27.83	59.93	49.45	0.1108
		Gmean_rej	3.45	3.45	3.45	89.39	65.17	74.78	67.95	87.59	77.92	0.1421
		NI_rej	13.20	9.86	12.23	79.35	41.55	41.96	41.67	80.30	69.91	0.1607
	Bayes	Bayes classifier	1.60	89.32	26.96	73.04	—	—	—	32.23	18.55	0.0313
		SMOTE	50.67	26.09	43.57	56.43	—	—	—	50.16	45.92	0.0719
		Cost-sensitive	51.30	10.35	39.46	60.54	—	—	—	65.99	56.81	0.1234
		Gmean_no_rej	30.48	30.12	30.38	69.62	—	—	—	69.38	57.19	0.1138
		NI_no_rej	35.56	26.50	32.93	67.07	—	—	—	66.49	55.84	0.1202
		Chow's reject	0.19	20.33	6.01	88.02	38.44	78.60	50.05	19.78	8.77	0.0742
		Gmean_rej	2.67	3.12	2.80	86.14	77.69	81.72	78.86	83.67	73.48	0.0822
		NI_rej	12.15	9.46	11.37	80.98	40.91	41.45	41.06	81.53	71.72	0.1786
Phoneme	kNN	kNN classifier	6.44	23.03	11.31	88.69	—	—	—	84.86	79.98	0.4261
		SMOTE	17.40	10.46	15.36	84.64	—	—	—	85.81	77.51	0.4106
		Cost-sensitive	13.69	11.55	13.06	86.94	—	—	—	87.37	79.90	0.4372
		Gmean_no_rej	14.09	11.00	13.18	86.82	—	—	—	87.43	79.86	0.4388
		NI_no_rej	12.53	12.77	12.60	87.40	—	—	—	87.32	80.28	0.4406
		Chow's reject	2.27	11.80	5.07	93.94	11.08	29.17	16.39	90.10	87.21	0.4683
		Gmean_rej	1.20	1.94	1.42	97.62	36.01	52.78	40.93	96.93	94.98	0.4509
		NI_rej	5.35	4.62	5.14	93.54	20.22	21.79	20.68	93.69	89.41	0.5086
	Bayes	Bayes classifier	10.44	31.25	16.55	83.45	—	—	—	78.46	70.92	0.2816
		SMOTE	24.61	12.53	21.06	78.94	—	—	—	80.70	71.02	0.3122
		Cost-sensitive	21.89	12.41	19.10	80.90	—	—	—	82.71	72.91	0.3248
		Gmean_no_rej	23.81	9.96	19.74	80.26	—	—	—	82.81	72.81	0.3331
		NI_no_rej	26.03	7.94	20.72	79.28	—	—	—	82.49	72.29	0.3359
		Chow's reject	1.17	13.14	4.69	93.05	20.36	62.33	32.68	80.01	75.37	0.2679
		Gmean_rej	0.17	0.14	0.16	99.25	78.47	91.71	82.36	96.97	95.24	0.1292
		NI_rej	10.05	4.35	8.37	88.92	23.50	28.33	24.92	90.32	82.69	0.3845
German	kNN	kNN classifier	9.67	69.03	27.48	72.52	—	—	—	52.75	40.24	0.0554
		SMOTE	49.80	21.39	41.28	58.72	—	—	—	60.30	53.23	0.0746
		Cost-sensitive	29.57	35.53	31.36	68.64	—	—	—	67.32	55.19	0.0888
		Gmean_no_rej	32.92	29.47	31.88	68.12	—	—	—	68.67	57.03	0.1017
		NI_no_rej	34.40	28.00	32.48	67.52	—	—	—	68.38	57.12	0.1037
		Chow's reject	2.17	35.53	12.18	81.03	27.40	55.60	35.86	43.97	30.40	0.0582
		Gmean_rej	6.20	6.07	6.16	80.66	68.63	72.53	69.80	78.48	67.65	0.0797
		NI_rej	17.30	16.83	17.16	73.89	34.85	33.50	34.45	73.63	63.57	0.1139
	Bayes	Bayes classifier	0.00	100.00	30.03	69.97	—	—	—	0.00	0.00	0.0000
		SMOTE	61.84	32.33	52.99	47.01	—	—	—	13.06	37.16	0.0183
		Cost-sensitive	21.43	39.27	26.78	73.22	—	—	—	69.06	57.64	0.1179
		Gmean_no_rej	32.23	26.73	30.58	69.42	—	—	—	70.23	58.96	0.1144
		NI_no_rej	30.80	29.50	30.41	69.59	—	—	—	69.12	57.87	0.1189
		Chow's reject	0.00	30.60	9.18	84.62	28.51	69.40	40.78	0.00	0.00	0.0567
		Gmean_rej	1.77	2.07	1.86	89.16	82.11	88.33	83.98	79.02	68.39	0.0689
		NI_rej	15.40	12.23	14.45	76.33	40.20	39.50	39.99	75.88	66.92	0.1334

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Data set	Method		$E_N(\%)$	$E_P(\%)$	$E(\%)$	$A(\%)$	$Rej_N(\%)$	$Rej_P(\%)$	$Rej(\%)$	$G(\%)$	$F(\%)$	NI
Diabetes	kNN	kNN classifier	13.56	48.33	25.69	74.31	—	—	—	66.72	58.29	0.1288
		SMOTE	42.46	17.96	33.91	66.09	—	—	—	67.45	62.80	0.1320
		Cost-sensitive	25.00	30.16	26.80	73.20	—	—	—	72.33	64.50	0.1495
		Gmean_no_rej	31.64	23.13	28.67	71.33	—	—	—	72.37	65.17	0.1538
		NI_no_rej	33.21	21.55	29.14	70.86	—	—	—	72.07	65.24	0.1580
		Chow's reject	5.16	26.35	12.56	81.61	24.00	46.48	31.84	68.65	60.00	0.1370
		Gmean_rej	5.79	4.32	5.28	86.66	64.01	68.46	65.57	85.55	78.96	0.1381
		NI_rej	15.28	11.60	14.00	79.25	33.38	33.13	33.29	79.81	73.62	0.1856
	Bayes	Bayes classifier	1.50	86.38	31.12	68.88	—	—	—	36.19	23.17	0.0501
		SMOTE	53.33	23.66	42.98	57.02	—	—	—	46.01	52.79	0.0861
		Cost-sensitive	19.16	41.87	27.08	72.92	—	—	—	68.51	59.95	0.1208
		Gmean_no_rej	28.76	23.66	26.98	73.02	—	—	—	73.57	66.29	0.1611
		NI_no_rej	37.72	16.96	30.48	69.52	—	—	—	71.37	65.56	0.1629
		Chow's reject	0.12	24.25	8.54	84.37	29.56	75.15	45.47	10.16	4.25	0.0827
		Gmean_rej	1.44	1.57	1.49	93.27	78.97	88.72	82.37	91.72	84.75	0.0971
		NI_rej	22.14	8.88	17.51	76.39	26.03	28.93	27.04	78.07	71.68	0.1893
Gamma	kNN	kNN classifier	5.49	35.11	15.90	84.10	—	—	—	78.31	74.15	0.3258
		SMOTE	27.12	15.56	23.06	76.94	—	—	—	77.88	72.23	0.2624
		Cost-sensitive	11.27	25.35	16.22	83.78	—	—	—	81.38	76.40	0.3251
		Gmean_no_rej	17.27	19.21	17.95	82.05	—	—	—	81.75	76.00	0.3102
		NI_no_rej	10.13	26.79	15.99	84.01	—	—	—	81.11	76.30	0.3275
		Chow's reject	2.14	22.93	9.45	88.79	11.31	23.87	15.72	82.58	79.83	0.3640
		Gmean_rej	1.01	3.88	2.02	95.61	55.00	52.61	54.16	94.74	93.82	0.3367
		NI_rej	3.95	10.84	6.37	91.21	27.28	28.59	27.74	89.57	87.01	0.3893
	Bayes	Bayes classifier	5.04	47.20	19.87	80.13	—	—	—	70.81	65.15	0.2348
		SMOTE	25.88	22.60	24.73	75.27	—	—	—	74.52	68.70	0.2183
		Cost-sensitive	13.38	31.31	19.68	80.32	—	—	—	77.13	71.05	0.2459
		Gmean_no_rej	21.82	22.45	22.04	77.96	—	—	—	77.83	71.23	0.2336
		NI_no_rej	12.02	33.29	19.50	80.50	—	—	—	76.59	70.62	0.2461
		Chow's reject	0.81	25.51	9.49	87.16	18.18	40.67	26.09	75.13	71.48	0.2634
		Gmean_rej	0.05	0.11	0.07	99.30	91.65	86.48	89.83	99.22	99.25	0.1070
		NI_rej	3.87	10.44	6.18	90.13	36.09	40.40	37.60	88.00	84.85	0.3066

TABLE 5

The “Equivalent” Costs and the Optimal Rejection Thresholds for Binary Class Data Sets

(a) kNN Classifier Based

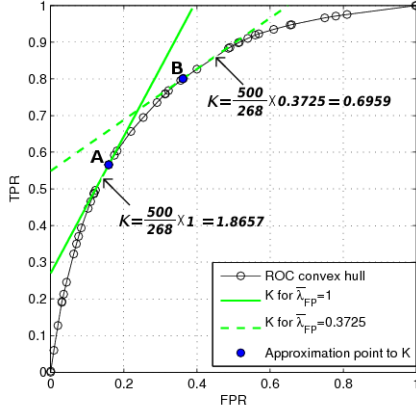
Data set	$\alpha_N^*(\bar{\lambda}_{FP})$	T_{rN}^*	T_{rP}^*	$\bar{\lambda}_{RN}$	$\bar{\lambda}_{RP}$
Ism	0.2312(0.0408)	0.0743(0.0085)	0.7643(0.0296)	0.0272	0.6616
Nursery	0.3482(0.0328)	0.1215(0.0565)	0.7125(0.0403)	0.0288	0.7914
Letter	0.3802(0.0321)	0.1284(0.0343)	0.6140(0.0517)	0.0760	0.4838
Rooftop	0.1372(0.0302)	0.0705(0.0610)	0.7733(0.0404)	0.0544	0.2823
Pendigits	0.4714(0.0771)	0.1745(0.0851)	0.3890(0.0555)	0.1710	0.1913
Optdigits	0.4061(0.0667)	0.1487(0.0372)	0.5913(0.0486)	0.0964	0.4481
Vehicle	0.1972(0.0769)	0.1101(0.0297)	0.6266(0.0719)	0.1045	0.1556
Yeast	0.3610(0.1333)	0.1245(0.0335)	0.5608(0.0536)	0.0937	0.3414
Phoneme	0.4651(0.0835)	0.1319(0.0180)	0.4543(0.0409)	0.1066	0.2985
German	0.3848(0.0785)	0.1915(0.0336)	0.6021(0.0368)	0.1542	0.3489
Diabetes	0.3725(0.0796)	0.1725(0.0455)	0.5284(0.0672)	0.1585	0.2398
Gamma	0.5682(0.0207)	0.1663(0.0225)	0.4188(0.0319)	0.1376	0.3103

(b) Bayes Classifier Based

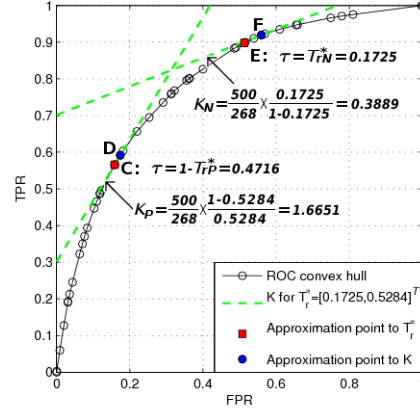
Data set	$\alpha_N^*(\bar{\lambda}_{FP})$	T_{rN}^*	T_{rP}^*	$\bar{\lambda}_{RN}$	$\bar{\lambda}_{RP}$
Ism	0.1420(0.0230)	0.0222(0.0168)	0.8560(0.0212)	0.0041	0.8198
Nursery	0.1052(0.0117)	0.0593(0.0076)	0.8845(0.0090)	0.0237	0.6242
Letter	0.1155(0.0066)	0.0687(0.0163)	0.8772(0.0128)	0.0273	0.6302
Rooftop	0.0786(0.0299)	0.0242(0.0080)	0.8363(0.0242)	0.0170	0.3147
Pendigits	0.4283(0.0472)	0.1521(0.0488)	0.6170(0.0394)	0.0782	0.5640
Optdigits	0.1883(0.0087)	0.1339(0.0047)	0.8274(0.0078)	0.0581	0.6240
Vehicle	0.2490(0.0262)	0.1301(0.0149)	0.5369(0.0384)	0.1287	0.1395
Yeast	0.4469(0.0734)	0.2668(0.0147)	0.6413(0.0137)	0.2093	0.4247
Phoneme	0.3364(0.0374)	0.1723(0.0266)	0.5511(0.0363)	0.1641	0.2115
German	0.4265(0.0137)	0.2802(0.0081)	0.6866(0.0085)	0.1736	0.5541
Diabetes	0.3808(0.0460)	0.2461(0.0174)	0.6638(0.0327)	0.2279	0.3020
Gamma	0.5859(0.0483)	0.1723(0.0212)	0.4660(0.0272)	0.1243	0.4028

Optimal values are listed as mean(standard deviation).

(a) Derived based on kNN classifier. (b) Derived based on Bayes classifier.



(a) For non-abstaining classification



(b) For abstaining classification

Fig. 3. Results on ROCCH of k NN for Diabetes. (a) For non-abstaining classification. (b) For abstaining classification.

reject option is added, the error rate may be reduced and the accuracy may be increased. But it is difficult to decide the rejection costs and the rejection thresholds for lack of information about the rejections. Regarding to *Chow's reject*, it is usually wasteful to reject lots of instances from the positive class with arbitrary settings on the rejection thresholds. On most data sets, $Gmean_rej$ achieves the highest accuracy and the lowest error rate of the positive class, at the price of considerably high reject rate. However, the accuracy of $Gmean_rej$ is lower than the conventional classifications on *Isr* and *Roofstop*. One explanation is that the goal of the G-mean based methods is to maximize the geometric mean of the accuracy within each class rather than the total accuracy. Compared with $Gmean_rej$ and *Chow's reject*, our NI_rej performs best on the whole with low error rate of the positive class, high accuracy, a certain amount of reject rate, high G-mean, high F-measure and the highest NI.

Table 5 lists the values of the optimal weight α_N^* and rejection thresholds T_r^* . The last two columns represent the “equivalent” rejection costs computed with the mean values of α_N^* and T_r^* . Moreover, these values are purely determined by the data sets besides the conventional classification algorithms. They can be adopted as “objective” references while the cost information is unknown. In addition, the “equivalent” costs of these data sets are consistent with human assumption, which also reflects the effectiveness of our NI based strategy.

Fig. 3 shows the ROC convex hull (ROCCH) of k NN for Diabetes generated from 90 validation sets by threshold averaging [44]. We just list some of the vertices in Table 6. And we use them to approximatively locate the parameters [15]. In Fig. 3a, we use equal and “equivalent” misclassification costs to compute the slopes, respectively. The slopes computed with costs are the same as those computed with rejection thresholds on ROCCH, so we only plot the latter in Fig. 3b. Point B in Fig. 3a lies between D and F. Due to approximation, points D and F that the slopes find are not cohere with the optimal

TABLE 6
Some ROCCH Vertices of k NN for Diabetes

Index (Point Label)	ROC Convex Hull Vertices (FPR, TPR)	Slope $\bar{K}$	Threshold τ
17	(0.1226, 0.4965)	2.1593	0.5106
18(A, C)	(0.1585, 0.5656)	1.9255	0.4514
19	(0.1588, 0.5662)	1.8502	0.4508
20(D)	(0.1739, 0.5915)	1.6778	0.4325
21	(0.1816, 0.6038)	1.5962	0.4211
...	...	...	...
28	(0.3554, 0.7958)	0.8326	0.2617
29(B)	(0.3615, 0.8002)	0.7357	0.2586
30	(0.3634, 0.8016)	0.6921	0.2578
31	(0.3997, 0.8265)	0.6881	0.2334
32	(0.4860, 0.8832)	0.6554	0.1781
33	(0.4901, 0.8858)	0.6543	0.1769
34(E)	(0.5129, 0.8983)	0.5475	0.1710
35	(0.5158, 0.8998)	0.5113	0.1701
36	(0.5389, 0.9104)	0.4605	0.1632
37(F)	(0.5596, 0.9196)	0.4434	0.1521
38	(0.5690, 0.9229)	0.3558	0.1345

(A~F: point labels shown in Fig. 3 are bolded)

threshold points C and E. In addition, the parameters can be adjusted with the graphical interpretations in Fig. 2 under the property P3 by the users.

5.4 Multi-Class Tasks

The detailed results on the multi-class data sets are shown from Table 7 to Table 9, including the performance evaluations and the values of the optimal parameters. Compared with the conventional classifications that have high error rates of the minority classes, *SMOTE* is effective in reducing these errors. *Cost-sensitive learning* classifies all instances to the class that has the minimum number of instances. Both $Gmean_no_rej$ and NI_no_rej perform well with low error rates of the minority classes, high G-mean and high NI. *Chow's reject* and $Gmean_rej$ reject lots of instances from the minority classes; besides, $Gmean_rej$ rejects lots of instances from the majority class. On the whole, our NI_rej performs best with low error rates of the minority classes, a certain amount of reject rate, high G-mean, and the highest NI.

In summary, the observations above suggest that:

TABLE 7
Results on Cardiocography and Thyroid

(a)

Data set		Method	$E_1(\%)$	$E_2(\%)$	$E_3(\%)$	$E(\%)$	$A(\%)$	$Rej_1(\%)$	$Rej_2(\%)$	$Rej_3(\%)$	$Rej(\%)$	$G(\%)$	NI
Cardiotocography	kNN	kNN classifier	2.39	39.49	28.67	9.71	90.29	—	—	—	—	74.88	0.4798
		SMOTE	6.71	23.35	21.26	10.22	89.78	—	—	—	—	82.37	0.5346
		Cost-sensitive	100.00	100.00	0.00	91.72	8.28	—	—	—	—	0.00	0.0000
		Gmean_no_rej	13.20	16.33	15.61	13.83	86.17	—	—	—	—	84.86	0.5105
		NI_no_rej	8.59	19.93	19.95	11.10	88.90	—	—	—	—	83.56	0.5224
		Chow's reject	0.53	19.66	15.47	4.42	94.96	5.70	40.54	27.28	12.32	80.56	0.4566
		Gmean_rej	0.39	4.88	5.09	1.40	98.10	19.23	68.77	31.41	27.11	91.05	0.4724
		NI_rej	4.44	9.46	15.96	6.09	92.94	13.24	25.08	6.34	14.31	88.04	0.5574
	Bayes	Bayes classifier	0.06	88.51	76.89	18.69	81.31	—	—	—	—	29.44	0.1579
		SMOTE	7.03	41.79	56.52	15.96	84.04	—	—	—	—	59.41	0.3378
		Cost-sensitive	100.00	100.00	0.00	91.72	8.28	—	—	—	—	0.00	0.0000
		Gmean_no_rej	17.28	15.81	22.01	17.47	82.53	—	—	—	—	81.49	0.4383
		NI_no_rej	12.13	22.10	27.06	14.75	85.25	—	—	—	—	79.13	0.4401
		Chow's reject	0.00	40.40	19.63	7.23	91.21	4.87	59.60	69.11	17.78	0.00	0.1158
		Gmean_rej	0.08	0.88	6.03	0.69	98.16	60.57	94.03	53.30	64.61	89.21	0.2510
		NI_rej	5.59	6.92	30.51	7.84	90.59	17.43	33.81	8.18	18.94	81.99	0.4666
Thyroid	kNN	kNN classifier	0.08	94.65	52.17	6.12	93.88	—	—	—	—	29.11	0.1726
		SMOTE	2.86	79.26	39.69	7.62	92.38	—	—	—	—	47.84	0.2245
		Cost-sensitive	100.00	100.00	0.00	97.69	2.31	—	—	—	—	0.00	0.0000
		Gmean_no_rej	25.69	41.03	22.54	26.41	73.59	—	—	—	—	69.65	0.2233
		NI_no_rej	5.85	70.92	32.28	9.78	90.22	—	—	—	—	56.64	0.2347
		Chow's reject	0.00	85.33	30.37	5.06	94.81	1.05	13.47	35.44	2.47	16.18	0.1277
		Gmean_rej	9.28	36.47	21.10	10.94	86.61	19.16	28.02	10.61	19.42	68.91	0.2413
		NI_rej	4.68	43.43	22.80	7.08	91.55	16.91	29.39	12.75	17.45	63.93	0.2555
	Bayes	Bayes classifier	0.06	99.24	87.04	7.14	92.86	—	—	—	—	7.24	0.0398
		SMOTE	0.89	95.49	78.46	7.51	92.49	—	—	—	—	18.74	0.0708
		Cost-sensitive	100.00	100.00	0.00	97.69	2.31	—	—	—	—	0.00	0.0000
		Gmean_no_rej	35.04	40.83	25.29	35.11	64.89	—	—	—	—	65.67	0.1885
		NI_no_rej	14.40	63.28	34.43	17.36	82.64	—	—	—	—	58.12	0.1921
		Chow's reject	0.03	97.29	71.09	6.64	93.30	0.30	1.90	20.35	0.84	8.40	0.0301
		Gmean_rej	2.80	4.62	11.29	3.09	86.06	77.82	85.94	43.99	77.45	76.66	0.1401
		NI_rej	8.67	31.50	29.03	10.31	85.80	29.45	42.05	11.42	29.68	60.96	0.1978

(b)

Cardiotocography	kNN	α_1^*	α_2^*	α_3^*
		0.2276(0.0592)	0.7138(0.1076)	1
		T_{r1}^*	T_{r2}^*	T_{r3}^*
	Bayes	0.0953(0.0391)	0.6001(0.1053)	0.7122(0.0888)
		α_1^*	α_2^*	α_3^*
		0.1332(0.0265)	0.5071(0.0697)	1
Thyroid	kNN	α_1^*	α_2^*	α_3^*
		0.1505(0.0395)	0.7065(0.1291)	1
		T_{r1}^*	T_{r2}^*	T_{r3}^*
	Bayes	0.0764(0.0269)	0.8058(0.0486)	0.8332(0.0450)
		α_1^*	α_2^*	α_3^*
		0.0367(0.0069)	0.4037(0.0751)	1
		T_{r1}^*	T_{r2}^*	T_{r3}^*
		0.0726(0.0171)	0.8996(0.0336)	0.9553(0.0219)

(a) Evaluation of the methods, “—”: not available, the best performance in each cell is bolded.

(b) Optimal weights and rejection thresholds are listed as mean(standard deviation).

1. Within the CFL category, both *SMOTE* and G-mean based methods are effective in the class imbalance problem. However, they are unable to process abstaining classifications.
2. Regarding to *Cost-sensitive learning*, it is feasible to apply the inverses of the class distribution ratios as the misclassification costs on binary class tasks. But on multi-class tasks, it may be ineffective. Moreover, the rejection costs are always hard to get.
3. *Chow's reject* would perform poorly if the rejection thresholds are arbitrarily given.
4. NI based strategy is a good choice for both non-abstaining and abstaining classifications. It can produce reasonable solutions on the minority classes.

6 CONCLUSION

In this paper, we propose a new strategy of CFL to deal with the class imbalance problem. Based on the specific property of mutual information that can distinguish different error types and reject types, we seek to maximize it as a general rule for dealing with binary/multi-class classifications with/without abstaining. A unique feature is gained in abstaining classifications when information is unknown about errors and rejects. To our best knowledge, no other existing approach is applicable to this scenario. Moreover, we can derive the “equivalent” costs for binary classifications. Generally, the “equivalent” costs will be changed accordingly to the distributions of the given data sets. Therefore, the present strategy provides an “objective” reference for CSL if users want

TABLE 8
Results on Car

(a)

Data set	Method		$E_1(\%)$	$E_3(\%)$	$E_4(\%)$	$E(\%)$	$A(\%)$	$Rej_1(\%)$	$Rej_3(\%)$	$Rej_4(\%)$	$Rej(\%)$	$G(\%)$	NI
Car	kNN	kNN classifier	0.50	35.76	62.03	7.82	92.18	—	—	—	—	66.65	0.6357
		SMOTE	7.13	29.08	52.41	9.25	90.75	—	—	—	—	73.23	0.6554
		Cost-sensitive	100.00	0.00	100.00	96.24	3.76	—	—	—	—	0.00	0.0000
		Gmean_no_rej	6.13	6.16	7.54	8.55	91.45	—	—	—	—	90.62	0.7091
		NI_no_rej	3.31	13.05	24.93	6.17	93.83	—	—	—	—	86.28	0.7356
		Chow's reject	0.00	1.45	1.45	0.23	99.70	5.70	78.47	97.10	24.37	50.00	0.4574
		Gmean_rej	0.05	4.33	4.06	0.94	98.84	11.44	23.64	42.90	21.50	95.47	0.6265
		NI_rej	1.94	9.70	20.87	4.35	95.34	6.66	0.30	2.46	7.00	88.68	0.7635
	Bayes	Bayes classifier	0.00	100.00	100.00	29.98	70.02	—	—	—	—	0.00	0.0000
		SMOTE	27.51	100.00	100.00	34.09	65.91	—	—	—	—	0.00	0.1733
		Cost-sensitive	100.00	0.00	100.00	96.24	3.76	—	—	—	—	0.00	0.0000
		Gmean_no_rej	26.48	17.07	12.75	24.06	75.94	—	—	—	—	80.44	0.4302
		NI_no_rej	26.05	15.40	21.88	23.25	76.75	—	—	—	—	79.38	0.4374
		Chow's reject	0.00	0.00	0.00	1.04	98.07	24.73	100.00	100.00	46.25	0.00	0.1828
		Gmean_rej	10.35	7.72	10.14	10.47	87.33	24.02	25.98	44.64	34.06	81.33	0.3486
		NI_rej	18.76	14.59	14.64	17.76	80.44	10.10	0.14	8.12	11.86	79.64	0.4388

(b)

kNN	α_1^*	α_2^*	α_3^*	α_4^*
	0.3029(0.0713)	0.5127(0.1216)	1.0183(0.1751)	1
	T_{r1}^*	T_{r2}^*	T_{r3}^*	T_{r4}^*
	0.2509(0.0514)	0.5751(0.0455)	0.7885(0.0604)	0.8027(0.0394)
Bayes	α_1^*	α_2^*	α_3^*	α_4^*
	0.0997(0.0072)	0.2825(0.0171)	0.9958(0.0984)	1
	T_{r1}^*	T_{r2}^*	T_{r3}^*	T_{r4}^*
	0.3083(0.0379)	0.7365(0.0340)	0.9209(0.0104)	0.9245(0.0115)

(a) Evaluation of the methods, “—”: not available, the best performance in each cell is bolded.

(b) Optimal weights and rejection thresholds are listed as mean(standard deviation).

to adjust the costs. For better understanding ROC curves in binary classifications, graphical interpretations of the theoretical ROC curve plots are explained in terms of the related parameters, such as cost terms and rejection thresholds. Empirical study confirms the advantages of the proposed strategy in solving class imbalance problems. At the same time, we recognize the disadvantage of the work that it will add an extra computational cost over the existing approaches. This difficulty will form a future work for advancing the study.

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TABLE 9
Results on Pageblock

(a)

Data set	Method		$E_1(\%)$	$E_3(\%)$	$E_5(\%)$	$E(\%)$	$A(\%)$	$Rej_1(\%)$	$Rej_3(\%)$	$Rej_5(\%)$	$Rej(\%)$	$G(\%)$	NI
Pageblock	k N N	kNN (k=5)	1.08	42.30	51.03	4.12	95.88	—	—	—	—	68.12	0.5466
		SMOTE	2.46	32.65	40.10	4.78	95.22	—	—	—	—	74.10	0.5666
		Cost-sensitive	100.00	0.00	100.00	99.49	0.51	—	—	—	—	0.00	0.0000
		Gmean_no_rej	6.04	9.26	24.48	7.08	92.92	—	—	—	—	84.74	0.5899
		NI_no_rej	3.73	15.30	26.66	5.37	94.63	—	—	—	—	82.17	0.6019
		Chow's reject	0.49	22.74	33.56	2.50	97.40	2.06	36.37	35.13	4.24	71.33	0.5022
		Gmean_rej	2.13	4.30	19.84	3.18	96.66	4.00	12.37	19.79	5.32	86.94	0.6028
		NI_rej	2.57	11.52	21.53	3.75	96.10	3.23	5.89	12.26	3.99	85.52	0.6179
	B a y e s	Bayes classifier	0.78	56.30	62.86	5.15	94.85	—	—	—	—	57.89	0.4559
		SMOTE	2.19	51.61	41.73	5.53	94.47	—	—	—	—	66.09	0.4982
		Cost-sensitive	100.00	0.00	100.00	99.49	0.51	—	—	—	—	0.00	0.0000
		Gmean_no_rej	14.45	17.06	16.23	14.77	85.23	—	—	—	—	84.66	0.4900
		NI_no_rej	13.71	17.33	15.47	14.22	85.78	—	—	—	—	82.33	0.5055
		Chow's reject	0.17	47.85	37.39	2.92	96.91	2.04	9.93	41.91	5.59	50.21	0.3362
		Gmean_rej	1.26	7.26	5.59	1.66	96.17	50.29	25.48	60.13	49.46	87.78	0.4714
		NI_rej	2.41	11.04	14.71	3.30	96.25	11.27	19.26	23.13	12.28	87.15	0.5749

(b)

kNN	α_1^*	α_2^*	α_3^*	α_4^*	α_5^*
	0.2234(0.0404)	0.6613(0.2007)	1.0869(0.2546)	0.9581(0.2774)	1
	T_{r1}^*	T_{r2}^*	T_{r3}^*	T_{r4}^*	T_{r5}^*
	0.1049(0.0608)	0.5775(0.1096)	0.8310(0.0667)	0.7229(0.1432)	0.7491(0.1046)
Bayes	α_1^*	α_2^*	α_3^*	α_4^*	α_5^*
	0.0219(0.0055)	0.4162(0.0428)	0.9210(0.2150)	0.7418(0.3106)	1
	T_{r1}^*	T_{r2}^*	T_{r3}^*	T_{r4}^*	T_{r5}^*
	0.0411(0.0125)	0.6620(0.1081)	0.9351(0.0254)	0.5595(0.1233)	0.7622(0.0907)

(a) Evaluation of the methods, “—”: not available, the best performance in each cell is bolded.

(b) Optimal weights and rejection thresholds are listed as mean(standard deviation).

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